Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

# Lecture 3A – Engineering Anthropometry I

EIN 3241: Human Factors and Ergonomics
Spring 2026

January 28, 2026

Dr. Jad Atweh

# Admin Issues

- Quiz 2
  - Wednesday February 4th

# Learning Objectives

- Understand the importance of anthropometry in engineering design
- Define key anthropometric terms and body dimensions
- Explore types of anthropometric data
- Review how anthropometric data is collected

# What is Anthropometry?

- Anthropometry: Anthro (man) – pometry (measurements)
  - Literal meaning: “measurement of humans”
- Anthropometry is:
  - The study and measurement of human body structures.
  - The study of human dimensions

**Design of workplaces**

Examples:

1. Workstation for standing or seated work
2. Machinery

**Design of consumer products**

Examples:

1. Automobiles
2. Furniture
3. Hand tools
4. Clothes

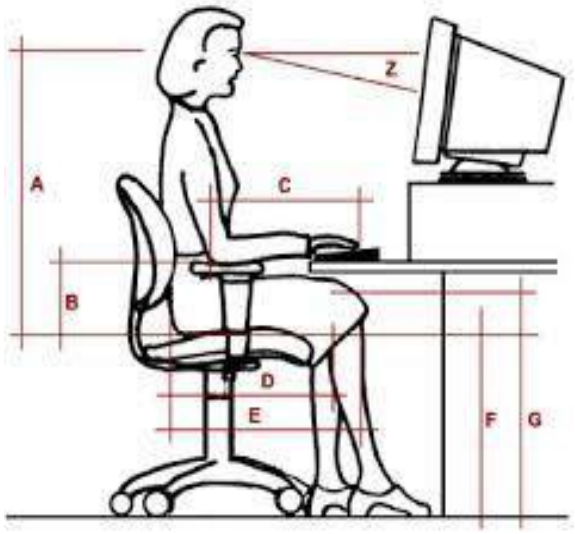
**Biomechanical modeling**

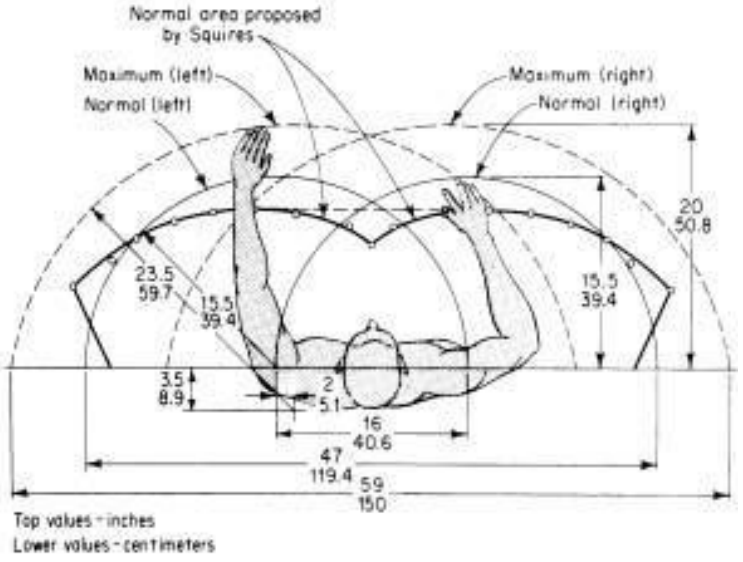
Example:

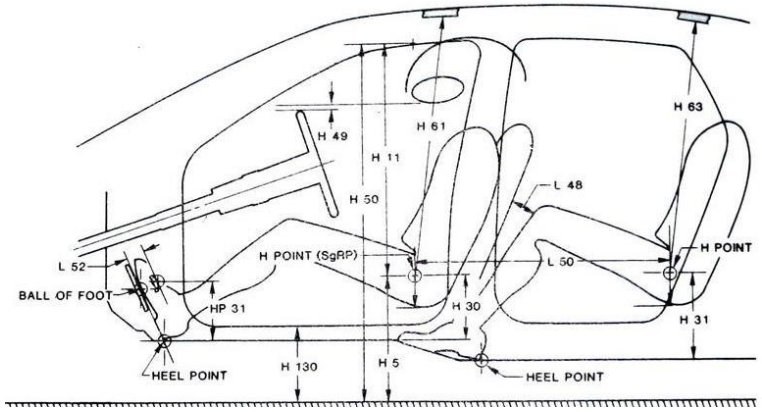
Stress imposed on workers' joints or muscles during work performance

# Design Applications

- Primary areas of application of anthropometric data are:
  - Clothing design
  - Workspace design
  - Design of equipment, tools, and machinery
  - Consumer product design



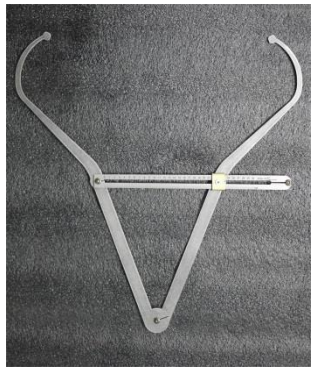



# Importance in Engineering Design

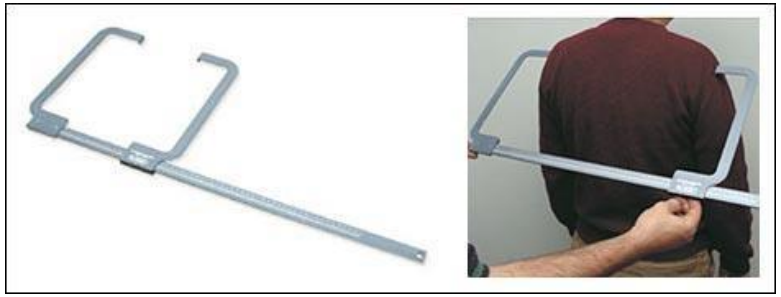
- Designer should accommodate body dimensions of the population that will be using the equipment
- Universal operability is always desired; at least 90% to 95% must be able to use the design
  - Universal operability can be achieved by adjustable designs.
- Certain aspects of design may call for:
  - Only one set of dimensions to be used.
  - Consider 5th, 50th, and 95th percentiles.
- Fit, Reach, Clearance, Posture & Performance…

# Measurement Devices and Techniques



Spread Caliper

Tape



Anthropometer

Sliding (Straight) Caliper

Dynamometer

Weight Scale

# Measurement Devices and Techniques

**Morant technique**:

- A set of grids is used to aid measurement.
- Grids are perpendicular to each other.
- Projections of body landmarks on grids are used as measurements.

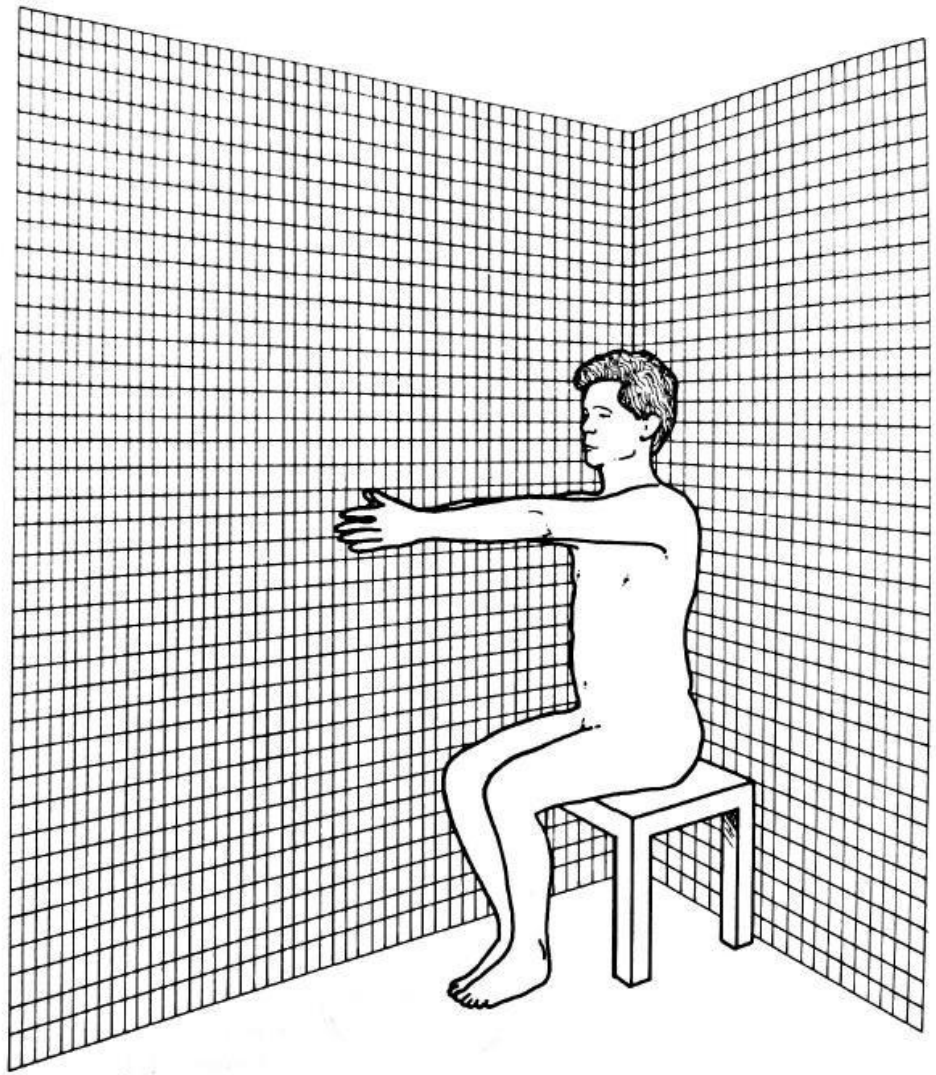


**Figure 5.2** Grid system.

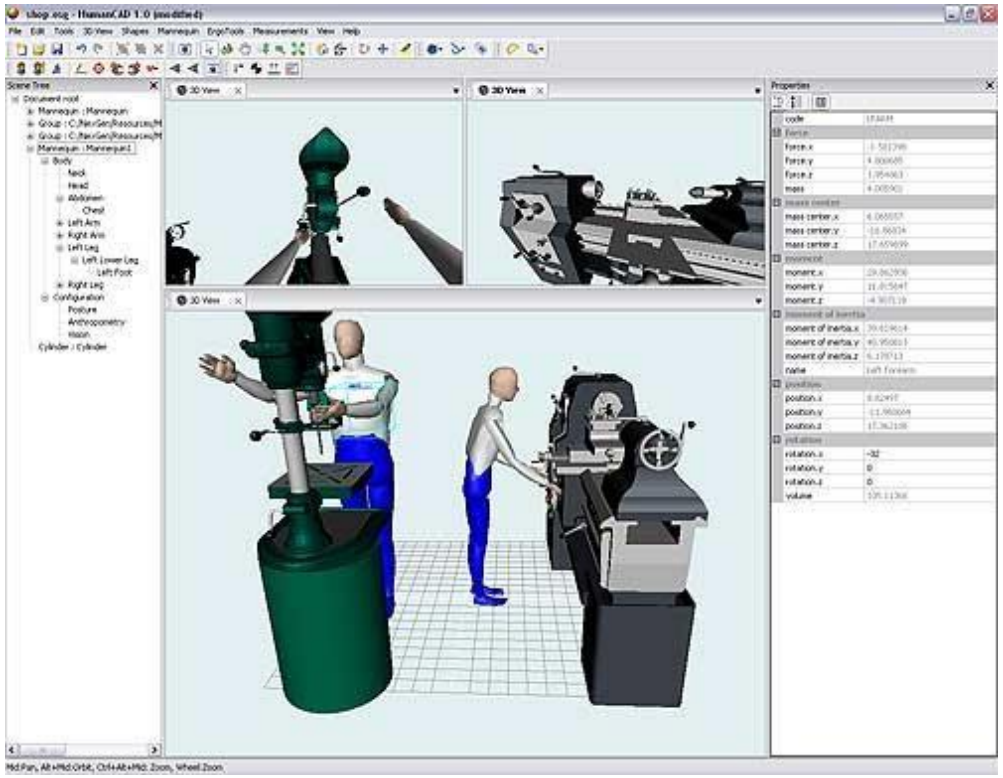
# Measurement Devices and Techniques

- Photographic/videotaping methods

# Measurement Devices and Techniques

Manikins:

- Manikins (2D or 3D) and computerized models can be used to simulate the anthropometric characteristics of a user population and evaluate  proposed designs.
  - –E.g. HumanCad Software



# Factors that affect anthropometric data

- Age:
  - Body size increases from birth to early/late 20s.
  - Stature reached at median age of 21.2 for males and 17.3 for females.
  - Body begins to shrink around 40, and more so with females

**TABLE 5.1** CANADIAN STATURE VARIATION BY AGE (CM)

| Age | Percentile 5th | Percentile 50th | Percentile 95th | Change from previous value (50th percentile) |
|---|---|---|---|---|
| 18–19 | 162.8 | 173.2 | 183.1 | |
| 20–24 | 162 | 173.3 | 184.2 | +0.1 |
| 25–29 | 159.8 | 173.5 | 187.9 | +0.2 |
| 30–34 | 161 | 172.9 | 184.9 | −0.6 |
| 35–44 | 159.2 | 171.7 | 184.4 | −1.2 |
| 45–54 | 156.9 | 169.7 | 182.9 | −2.0 |
| 55–64 | 153.9 | 167.6 | 181.1 | −2.1 |
| >64 | 153.9 | 165.4 | 177.3 | −2.2 |

*Source:* Adapted from Ref. [2].

# Factors that affect anthropometric data

- Gender:
  - Men are in general larger than women at any given percentile and body dimensions except for hip and thigh area.
  - Gender differences also varies between ethnic groups.
- Ethnic Background

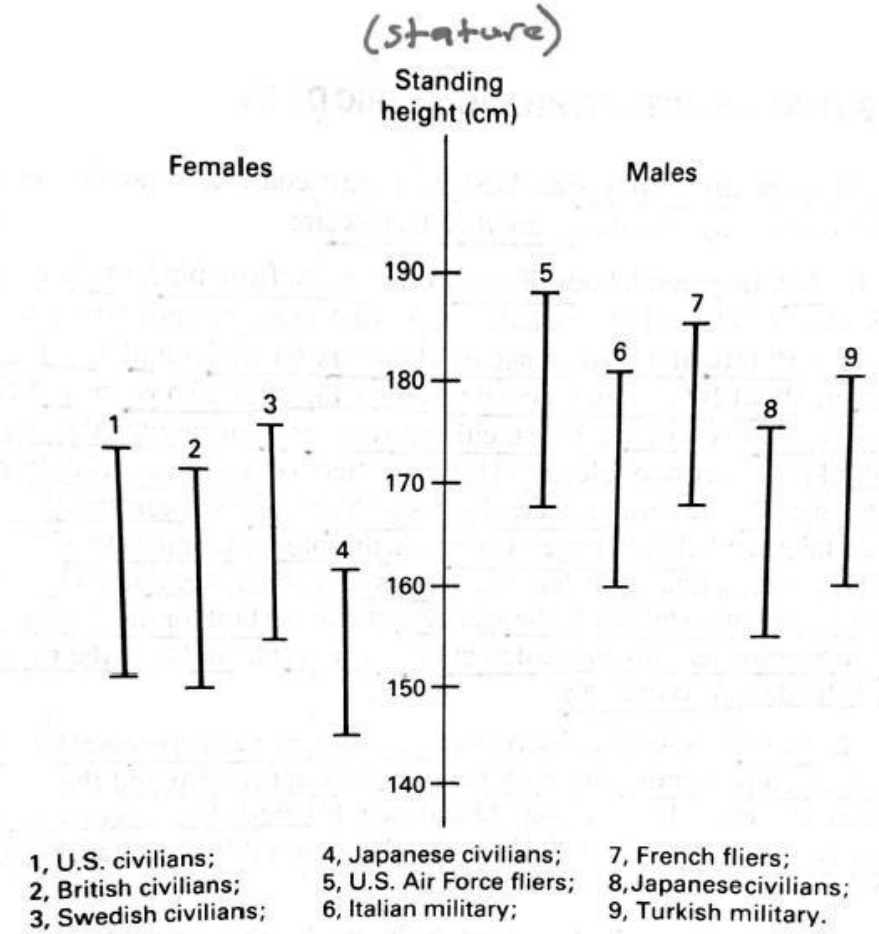



**Figure 5.3** Human variability ranges (5th and 95th percentiles). (Reproduced with permission of McGraw-Hill, Inc. From *Human Factors in Engineering and Design, 5/e*, Sanders and McCormick, 1982.)

# Factors that affect anthropometric data

- Body position
  - Posture affects body size
  - Slumping is one of the major reasons for the variability in height measures (affects by about 2 cm)
- Clothing
  - Adds to body size e.g. shoes , gloves, jacket.
  - “Furthermore, all anthropometric measurements are taken in nude.”

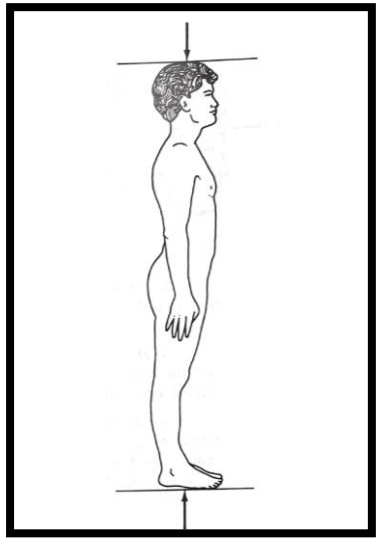
# Two Categories of Anthropometric Data

- Structural dimensions (static anthropometry)
  - Taken with the body in various standard and still positions.
  - Deals with simple dimensions of the stationary human being.
  - E.g., weight, stature or height of the subject during standing, standing eye height, standing shoulder height, standing elbow height, chest depth, abdominal depth…
- Functional dimensions (dynamic anthropometry)
  - Taken during physical activity (e.g., crawling height in mines)
  - Photographic/videotaping methods.
  - E.g., reach zones during standing, forward grip reach, vertical grip reach, side grip reach…

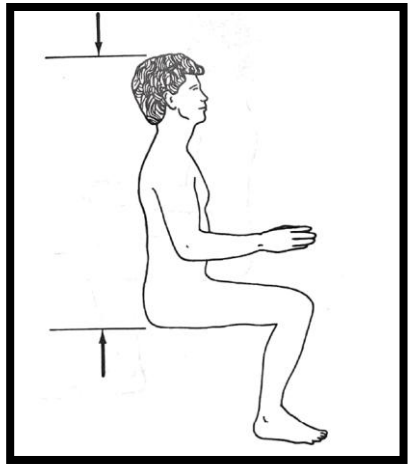
# A Few Anthropometric Measures

| No | Subject # | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Name** | | | | | | | | | | | | |
| 1 | Gender (F/M) | | | | | | | | | | | | |
| 2 | Age (year) | | | | | | | | | | | | |
| 3 | Weight (kg) | | | | | | | | | | | | |
| 4 | Stature (cm) | | | | | | | | | | | | |
| 5 | Eye Height, Standing (cm) | | | | | | | | | | | | |
| 6 | Acromial Ht., Standing (cm) | | | | | | | | | | | | |
| 7 | Knuckle Ht., Standing (cm) | | | | | | | | | | | | |
| 8 | Sitting Height (cm) | | | | | | | | | | | | |
| 9 | Knee Height, Sitting (cm) | | | | | | | | | | | | |
| 10 | Popliteal Height, Sitting (cm) | | | | | | | | | | | | |
| 11 | Elbow Rest Ht., Sitting (cm) | | | | | | | | | | | | |
| 12 | Hip Breadth, Sitting (cm) | | | | | | | | | | | | |
| 13 | Elbow-to-Elbow Breadth (cm) | | | | | | | | | | | | |
| 14 | Thigh Clearance, Sitting (cm) | | | | | | | | | | | | |
| 15 | Eye Height, Sitting (cm) | | | | | | | | | | | | |
| 16 | Arm Reach (cm) | | | | | | | | | | | | |
| 17 | Hand Length (cm) | | | | | | | | | | | | |
| 18 | Foot Length (cm) | | | | | | | | | | | | |
| 19 | Chest Depth (cm) | | | | | | | | | | | | |

# Static Anthropometry (Structural Dimensions)

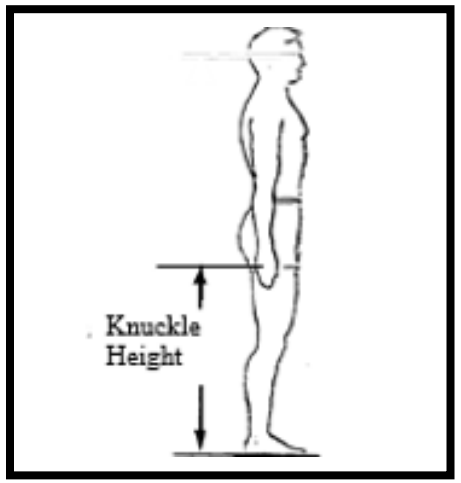


**Stature:** With the subject standing erect and looking straight ahead, this is the measure taken from the standing surface to the top of the head. Add 2.5 cm (1 in.) for men's shoes, 2 cm (0.8 in.) for caps, up to 8 cm (3.1 in.) for women's shoes, and 3.8 cm (1.5 in.) for steel helmet.

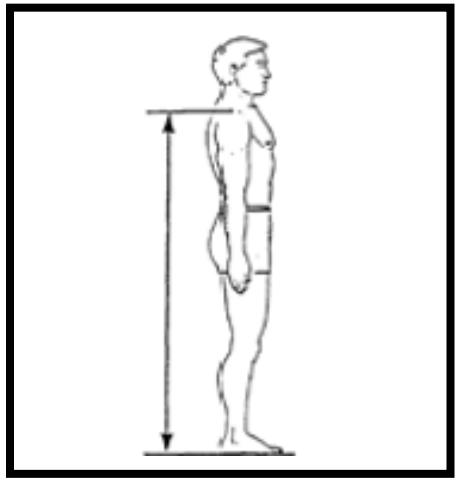


**Sitting Height:** This is the vertical distance between the seat surface and the top of the head with the subject sitting erect, looking straight ahead and the knees at the right angles. For heavy clothing, add 1.2 cm (0.5 in.); add 2 cm (0.8 in.) for caps) and 3.8 cm (1.5 in.) for steel helmets.

# Static Anthropometry (Structural Dimensions)

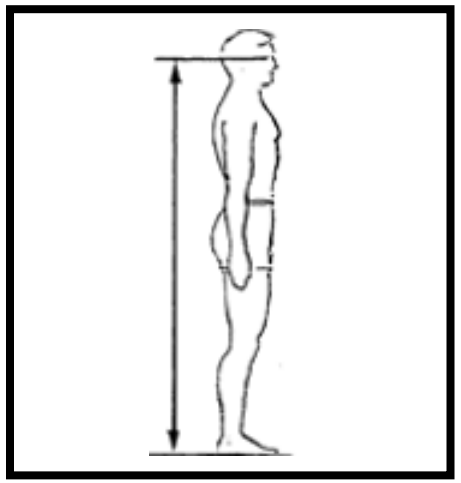



**Knuckle Height:** The vertical distance from the floor to the knuckle (typically the third joint) with arms hanging naturally at the sides in a relaxed standing position.



**Acromial Height:** The vertical distance from the floor to the acromion process (the bony tip of the shoulder) when standing upright.

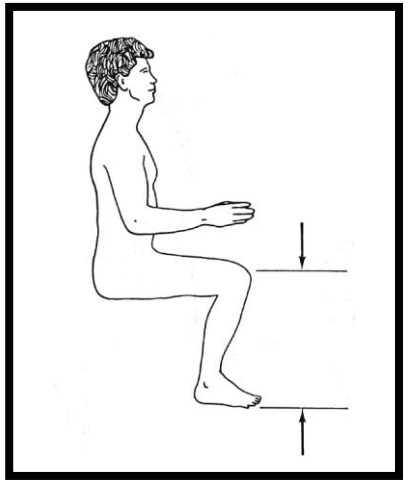
# Static Anthropometry (Structural Dimensions)



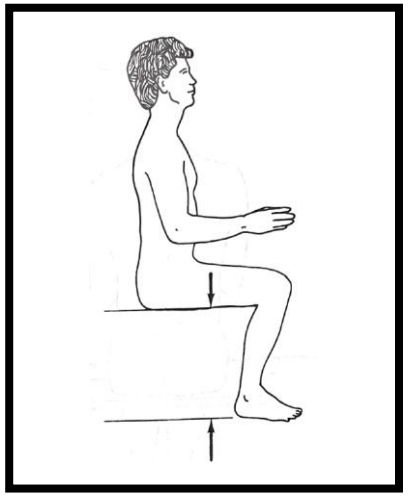
**Eye Height:** The vertical distance from the floor to the inner corner of the eye (canthus) when standing erect

**Eye Height (Sitting):** The vertical distance from the sitting surface to the inner corner of the eye while seated with erect posture.

# Static Anthropometry (Structural Dimensions)

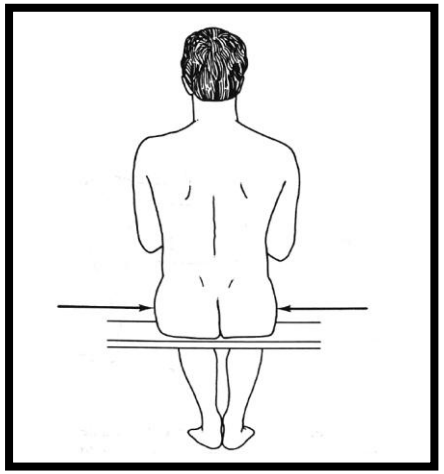


**Knee Height:** Vertical distance is measured from the floor to the upper most point on the knee. For light clothing and men's shoes, add 3 cm (1.2 in.); for boots add 4.5 cm (3 in); for women shoes and light clothing, add up to 8 cm (3.1 in).

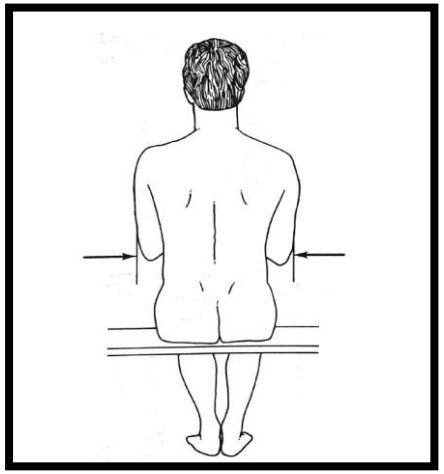


**Popliteal Height:** This is the vertical distance from the floor to the underside of the thigh next to the knee with the subject sitting erect, knees at right angles,and the bottom of the thigh and the back of the knee barely touching the seat pan. Clothing adjustments are similar to knee height data.

# Static Anthropometry (Structural Dimensions)

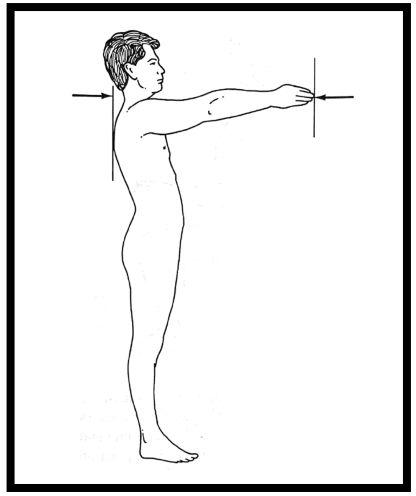


**Hip breadth, sitting :** This is the maximum horizontal distance across the hips with the subject sitting erect and knees together. Add 1.5 cm (0.6 in.) for light clothing and 4 cm (1.6 in.) for heavy clothing.

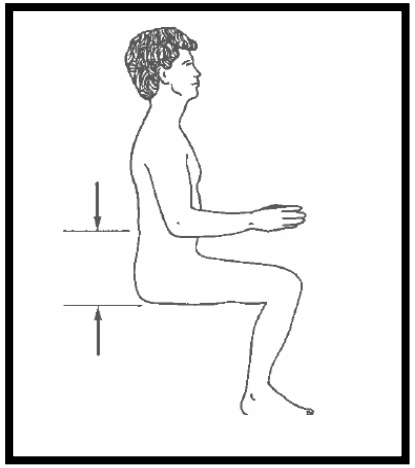


**Elbow-to-elbow breadth, sitting:** This is the horizontal distance between the lateral surfaces of the two elbows. The subject sits erect, hips and knees together, upper arms vertical and touching the sides of the body lightly. Add 1.5 cm (0.6 in.) for light clothing and 12 cm (4.8 in.) or more for heavy clothing.

# Static Anthropometry (Structural Dimensions)

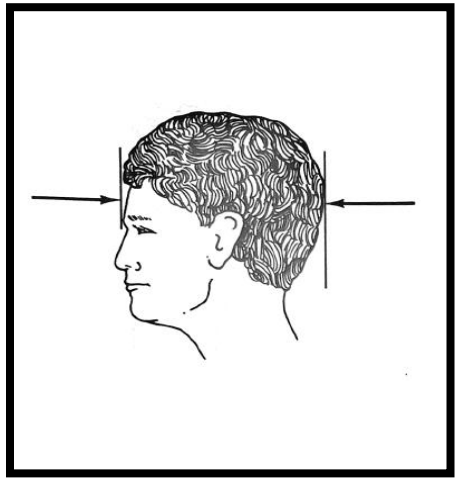


**Arm reach:** Horizontal distance is measured between the tip of the middle finger and the posterior surface of the right shoulder. Subject stands erect with ankles and knees together. The arm is extended to its maximum. Add 1.5 cm (0.6 in.) for light clothing and light gloves. Add 2 cm (0.8 in.) or more for heavy clothing and gloves. For manipulation by fingers, subtract 8 cm (3.2 in.); for whole hand manipulation, subtract 12 cm (4.8 in.).

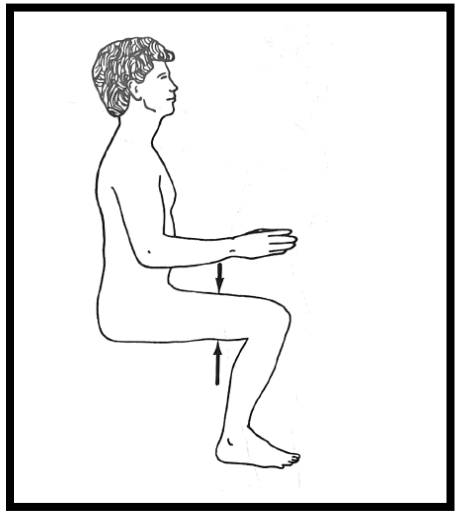


**Elbow rest height (Sitting):** Vertical distance from the seat surface to the underside of the elbow.

# Static Anthropometry (Structural Dimensions)

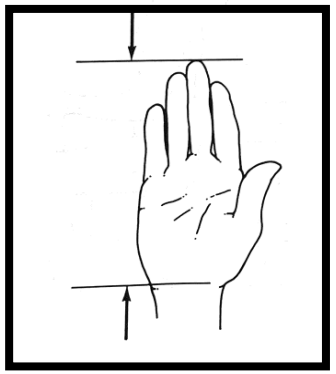


**Head length:** This is the maximum length of the head from the most front to the most rear of the head. Adjustment must be made due to the type of head gear worn.

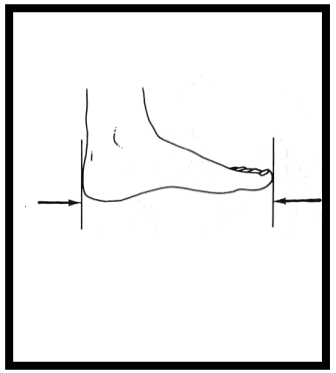


**Thigh clearance (sitting) :** This the vertical distance between the seat surface and the point of intersection of the thigh with the abdomen. Add 0.5 cm for light clothing and 3.5 cm or more for heavy clothing.

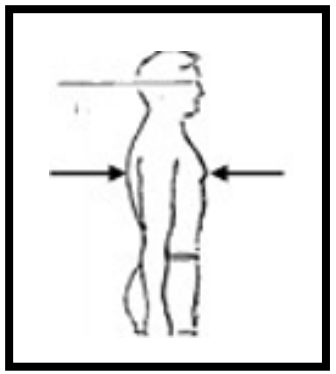
# Static Anthropometry (Structural Dimensions)



**Hand length:** This is the distance from the base of the thumb to the tip of the middle finger of the right hand. The hand is extended straight on the arm. Add 0.4 cm (0.16 in.) for light gloves, and 0.8 cm (0.32 in.) or more for heavy gloves.



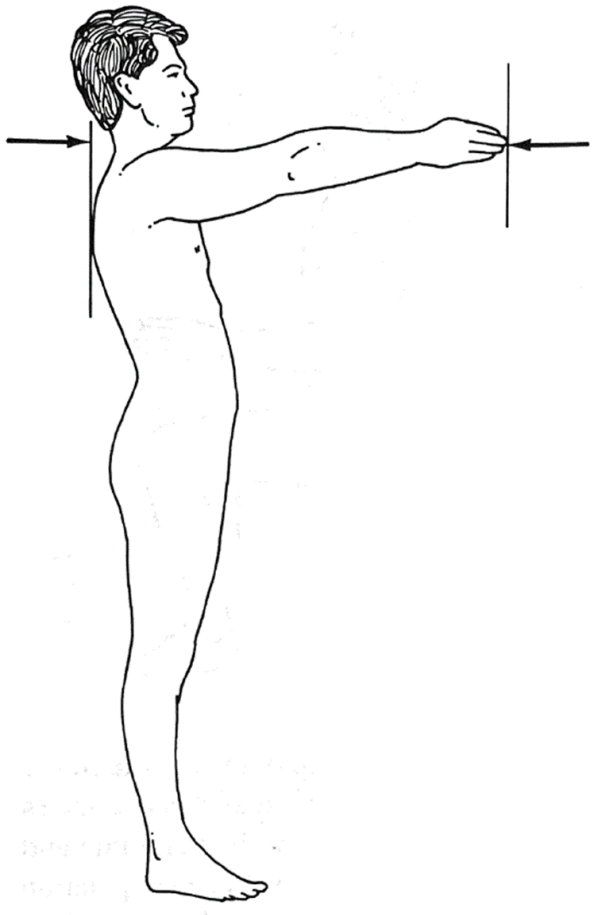
**Foot length:** This is the horizontal distance from the back of the heel to the tip of the longest toe. Subject stands erect with his weight equally distributed on both feet. Add 3 cm (1.2 in.) for men's shoes, 3.7 cm (1.5 in.) boots, and 2 cm (0.8 in.) for women's shoes.



**Chest Depth (Anterior–Posterior Thorax Depth):** The horizontal distance from the most anterior point on the chest (typically the sternum) to the most posterior point on the back (usually the spine/scapula) at the same vertical level.

# What did we learn today?

- What is anthropometry
- Design applications
- Measurement devices and techniques
- Factors that affect anthropometric data
- Structural dimensions…



# Next Time…

- Human Variability & Correlations of Body Measurements
- Calculating Percentiles…
- Anthropometry Process…
- Anthropometry Principles….

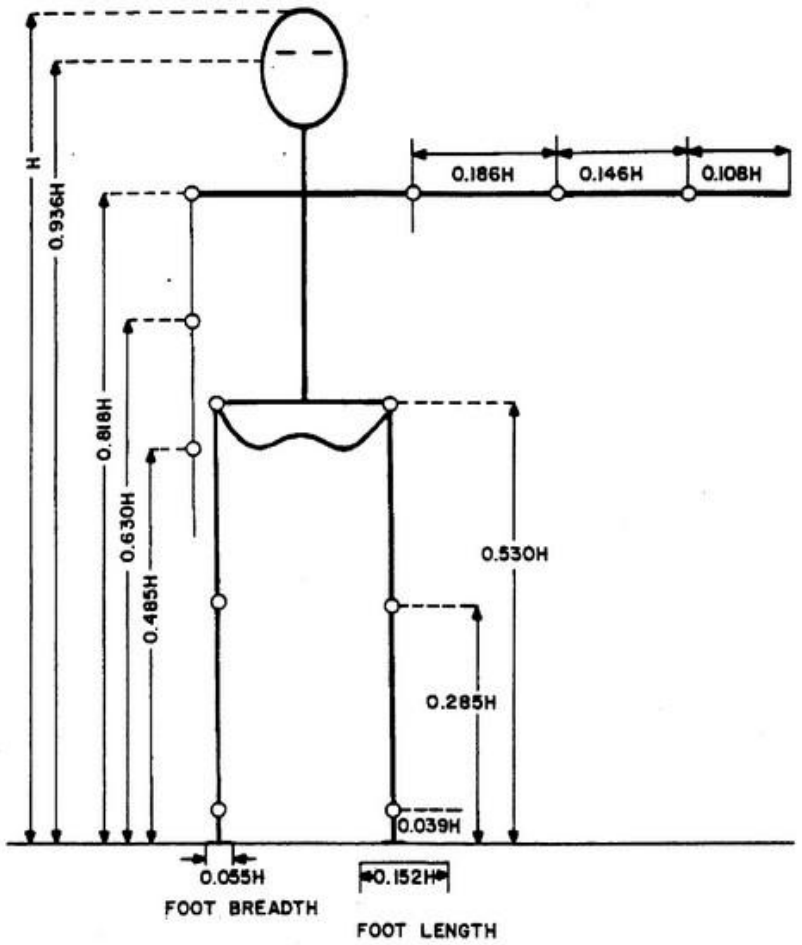



**Figure 3.9** Some body segment lengths expressed as proportion of body stature by Drillis and Contini (Roebuck, Kroemer, and Thomson, 1975).

UF

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LEADING THE CHARGE, CHARGING AHEAD